AI and PolicyMap in Public Health

In class assignment. Complete each part and hand in the finished document.

Name	Date	County and state you chose

How this assignment works

You will build a community health report for a county of your choice in PolicyMap, ask Gemini to evaluate it, check Gemini's work, fill a gap using free public data, and cite your sources in APA style. Short answers are fine. Where you see a box, write in your own words. There are no trick questions, and honest observations about what worked and what did not are exactly what I am looking for.

Before you start

Sign in to Gemini with your campus Google account.

Never paste personal or identifiable health information about real people into any AI tool. County level public data like today's is fine.

Keep your Gemini chat open. You will need it for the verification and citation steps.

Part 1. How generative AI works

How an AI model is like a brain

Neurons and synapses. A large language model is a neural network. It is made of simple artificial neurons joined by billions of weighted connections, which work roughly like synapses.

Associations. Words are turned into numbers that sit close to related words. The model learns that nurse goes with hospital and patient.

Learning. During training the model guesses the next word billions of times. Connections that lead to good guesses get stronger, much as practice strengthens synapses in your brain.

The limit. The model predicts what usually comes next. It has no way to know whether that prediction is true.

Not a calculator. A simple math program is deterministic. If A = 3 and B = 5, then C = A + B gives 8 every time you run it. An LLM is not. Ask it the same question twice and you get similar answers, but never exactly the same one.

1a. The prediction game. Write your prediction for each sentence as your instructor reads it.

Round	Sentence start	Your prediction	Did the class agree?
1	*Twinkle, twinkle, little star, how I wonder…*		
2	*The biggest health problem in our community is…*		
3	*I, Nephi, said unto my father…*		

> **From predicting to problem solving**
>
> **Then.** Early AI predicted an answer in one pass. Ask it for a county health statistic and it wrote the most likely looking number, whether or not that number was real.
>
> **Next.** Chat assistants were trained to follow instructions. They give a better answer in one try, sometimes after a quick web search, but they do not check their own work.
>
> **Now.** Reasoning agents like Gemini Deep Research work in a loop. They make a plan, search, read what they find, notice a dead end, change their search terms, and try again.
>
> **Picture a maze.** A prediction model draws the whole path in one guess. An agent walks it, backs up from dead ends, and tries another branch.
>
> **Still predicting.** Underneath, every step is still prediction. The loop makes AI more persistent, not more truthful, so you still trace claims back to the original.

1b. Using the brain comparison and the prediction game, explain in your own words how a tool like Gemini produces an answer. What did round 3 teach you? Why do you still need to check the work of an agent like Deep Research?

1c. The sliding scale of subjectivity. The more subjective a question is, the more critical you need to be of the answer. Use the scale on the resource page at the end of this assignment. Add two public health questions of your own. Place each one on the scale and say how you would check the answer.

A question you might ask AI	Where on the scale?	How would you check the answer?
How is measles transmitted?	*Objective*	*Compare with the CDC measles page. Low risk, but still check any numbers.*

A question you might ask AI	Where on the scale?	How would you check the answer?

Part 2. Build your county report in PolicyMap

1. Open PolicyMap from the library's database list and sign in with your campus account if asked.
2. Click **Reports** at the top of the screen.
3. Choose **Community Health Report** if it is available. If it is not, choose **Community Profile**.
4. Type your county, for example **Latah County, ID**, and select it from the list.
5. Click to generate the report and read through it.
6. Save the report as a PDF using the download or print option. If a print window opens, set the destination to **Save as PDF**.

Report type you used	Date you generated it

2a. List three things in the report that caught your attention or surprised you.

Part 3. Let Gemini interview you

Good prompts often start by letting the AI ask *you* questions. Upload your PDF to Gemini (click the **+** or paperclip in the prompt bar), paste the prompt below, and answer Gemini's questions.

> **Prompt 1. The interview**
>
> I am a public health student. I want to use the attached PolicyMap report for [YOUR COUNTY, STATE] to understand the health of this community and identify where the data falls short.
>
> Before you do anything else, interview me. Ask me one question at a time about my purpose, my audience, and what a useful evaluation would look like to me. Stop after no more than five questions, then summarize what you learned and wait for me to say go.

3a. Which question from Gemini was most useful, and how did answering it change what you asked for?

Part 4. Have Gemini evaluate your report

After the interview, paste this prompt into the same chat.

> **Prompt 2. The evaluation**
>
> Using what you learned from our interview, evaluate the attached community health report for [YOUR COUNTY, STATE]. Organize your answer in four parts.
>
> **1. Strengths.** What does this report do well? Which indicators are most useful for understanding community health?
>
> **2. Missing data.** What data points would a public health professional expect in a community health assessment that are missing or thin here? Think about health outcomes, health behaviors, access to care, social determinants of health, environment, and subgroup differences.
>
> **3. Cautions.** Where could these numbers mislead someone? Consider small populations, margins of error, how old the data is, and anything unusual about this county.
>
> **4. Where to look.** For each missing item, name a free, reputable public source where I could find it.
>
> Rules. Only report numbers that actually appear in the PDF and tell me the page or section where each appears. Label anything that comes from your general knowledge rather than the document as OUTSIDE THE REPORT.

4a. Record the most useful parts of Gemini's answer in your own words.

Two strengths of the report	Three missing data points	One caution about the data

Part 5. Make the AI check itself

AI tools sound equally confident whether they are right or wrong. Ask Gemini to audit its own answer, then check two of its numbers yourself against the PDF.

Prompt 3. The self check

Now check your own work. List every number and factual claim you made in your last answer. For each one, tell me whether it came from the PDF (with the page or section) or from outside the report, and rate your confidence as high, medium, or low. Flag anything you cannot locate in the document, and correct any mistakes you find.

5a. Pick two numbers or claims from Gemini's answer and check them against your PDF.

Number or claim Gemini gave	What the PDF actually says (page or section)	Match?

5b. Did Gemini catch any of its own mistakes? Did you find any it missed?

Part 6. Fill one gap with free public data

Pick one missing data point from Part 4 and find it on the free web. Use the source list and search tips on the resource page. You may ask Gemini where to look with the prompt below, but open every link yourself and get the number from the original source, not from the AI.

Optional prompt. Where should I look?

I need current, free, reputable public health data for [YOUR COUNTY, STATE] on [TOPIC, for example adult obesity, uninsured rates, or overdose deaths]. Suggest the three best sources, explain what each one measures, the most recent year it covers, and the smallest geography it reports. Give me the web address for each. If you are not certain a source still exists or has county level data, say so.

What you looked for	Value you found	Year of the data	Source (organization and web address)

6a. How did you decide this source was trustworthy? Think about who publishes it, how current it is, and how they collected the data.

Part 7. Document your sources in APA style

Every number you use in public health work should trace back to a source someone else can check. AI can format APA citations quickly, but it often invents details such as authors, dates, and DOIs when it is asked to recall a source. The safe approach is to give the AI the details yourself and ask it only to format them.

> **Prompt 4. Format an APA citation**
>
> Create an APA 7th edition reference list entry and a matching in text citation for the source below. Use only the details I give you. Do not invent an author, date, DOI, or web address. If something APA requires is missing, tell me what it is instead of guessing.
>
> Source details. Title [ ]. Author or organization [ ]. Publication or update date [ ]. Web address [ ]. Date I accessed it [ ]. Type of source, for example data report, web page, or AI chat [ ].

Check every AI generated citation

☐ The author or organization, title, and date match the actual source.

☐ The web address opens the right page. Any DOI works at doi.org.

☐ The title of a report or web page is in *italics* and in sentence case.

☐ Nothing appears that you did not give the AI.

Example formats

These show the pattern. Replace the bracketed parts with your real details.

- PolicyMap. ([Year]). *[Report title as shown on the report]* [Data report]. Retrieved [Month Day, Year], from [web address]
- [Organization]. ([Year]). *[Page or report title]*. [Web address]
- Google. ([Year, Month Day]). *[Short title describing your chat]* [Google Gemini chat]. Gemini. [Share link, if your account allows one]

Campus Gemini accounts may not create a share link. In that case leave the link off and paste your prompts into the appendix box at the end, as APA suggests.

7a. Write APA references for the three sources you used today. Your PolicyMap report, the source from Part 6, and your Gemini chat.

1. PolicyMap report

2. Free web source from Part 6

3. Gemini chat

Part 8. Reflection

This part is open ended. Write what you actually think. A short paragraph for each is plenty.

8a. What did you learn today about using AI in public health? What was it good at, and where did it fall short?

8b. When is it appropriate to use AI in public health work or in general, and when is it not? Consider accuracy, privacy, bias, and who is affected by the decisions the data supports.

8c. Name one habit you will carry forward when you use AI for school or work.

Appendix. Paste your Gemini prompts here if your chat has no share link (optional otherwise).

Resource page. Keep this for later

The sliding scale of subjectivity

More objective	In between	More subjective
How does glycolysis work? What is the formula for BMI? How is measles transmitted?	Which programs best reduce teen vaping? Why is this county's uninsured rate high? Which indicators matter most for this community?	Should schools require vaccines for entry? Should soda be taxed? The Israeli and Palestinian conflict
AI is usually reliable. Still verify specific numbers and citations.	**Be moderately critical.** Ask for evidence and for competing explanations.	**Treat the answer as one viewpoint among many.** Ask for multiple stakeholder perspectives and read them yourself.

Numbers are a special case. Even on objective topics, AI frequently gets statistics, dates, and citations wrong. Always check them against the original source.

Checklist for verifying AI output

- ☐ Ask the AI where each claim came from, and open the source yourself.
- ☐ Ask the AI what part of its answer it is least sure about.
- ☐ Check every number against the original document or dataset.
- ☐ Ask for the strongest opposing view on anything subjective.
- ☐ Check the date. AI training data and many datasets lag by a year or more.
- ☐ Compare with a second source that is not an AI.

Free, reputable public health data

Source	What you will find	Where
County Health Rankings & Roadmaps	Rankings and dozens of county health measures, each with its original source listed	countyhealthrankings.org
CDC PLACES	Estimates of chronic disease, health behaviors, and prevention for counties, cities, and census tracts	cdc.gov/places
CDC WONDER	Births, deaths, and causes of death you can query by county and year	wonder.cdc.gov
U.S. Census Bureau	Population, age, income, poverty, insurance, and housing from the American Community Survey	data.census.gov

Source	What you will find	Where
CDC/ATSDR Social Vulnerability Index	Which neighborhoods are most vulnerable during emergencies	atsdr.cdc.gov (search Social Vulnerability Index)
USDA Food Environment Atlas	Food access, grocery stores, and food assistance	ers.usda.gov
HRSA Data Warehouse	Health professional shortage areas and health center locations	data.hrsa.gov
Healthy People 2030	National targets to compare your county against	odphp.health.gov/healthypeople
Idaho Department of Health and Welfare	State health statistics, vital records, and survey data	healthandwelfare.idaho.gov
Local health districts and hospitals	Community health assessments. Nonprofit hospitals must publish a community health needs assessment every three years	Search the county name plus "community health needs assessment"

Search tips

- Add **site:.gov** or **site:.edu** to a search to limit results to government or university sites.
- Search **"community health needs assessment" "Latah County" filetype:pdf** to find local reports.
- Read laterally. When you land on an unfamiliar site, open a new tab and look up who runs it before you trust it.
- Check the data year, the geography, and the method. Small counties often have large margins of error.
- Watch for local quirks. A college town like Moscow or Rexburg has many students, which shifts age and income numbers.